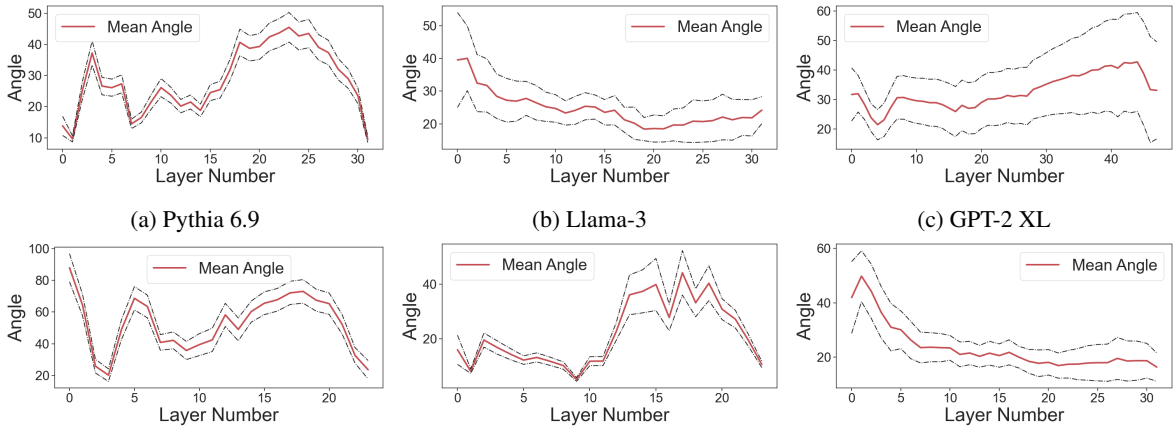
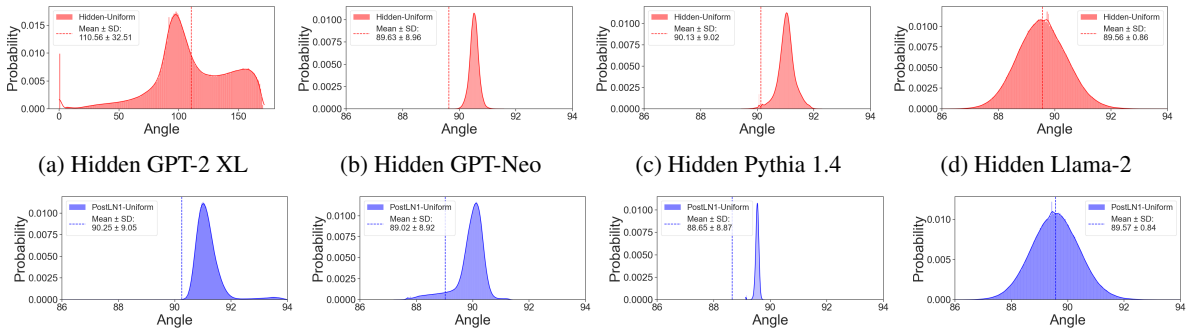


(a) Pythia 6.9 (b) Llama-3 (c) GPT-2 XL

(d) GPT-Neo (e) Pythia 1.4 (f) Llama-2

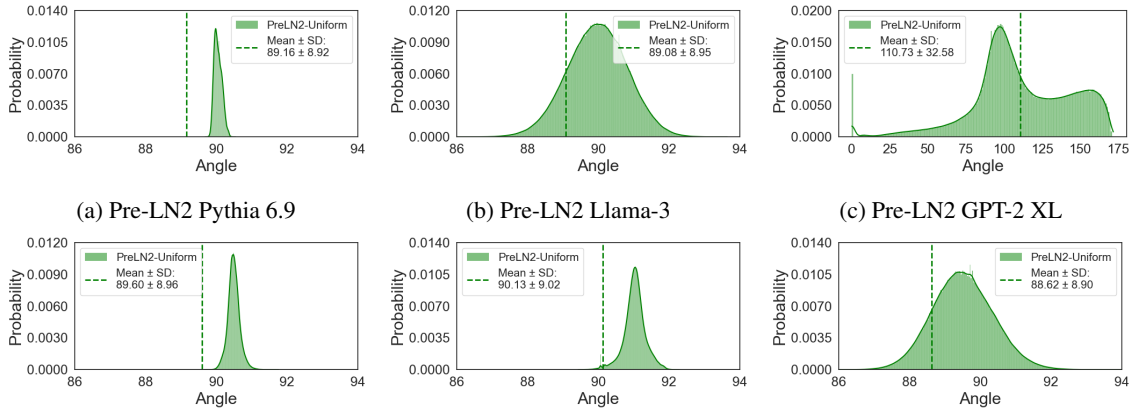
Figure 10: Rotation angle (in degrees) between the hidden vectors and Post-LN2 vectors across all layers for all models



(a) Hidden GPT-2 XL (b) Hidden GPT-Neo (c) Hidden Pythia 1.4 (d) Hidden Llama-2

(e) Post-LN1 GPT-2 XL (f) Post-LN1 GPT-Neo (g) Post-LN1 Pythia 1.4 (h) Post-LN1 Llama-2

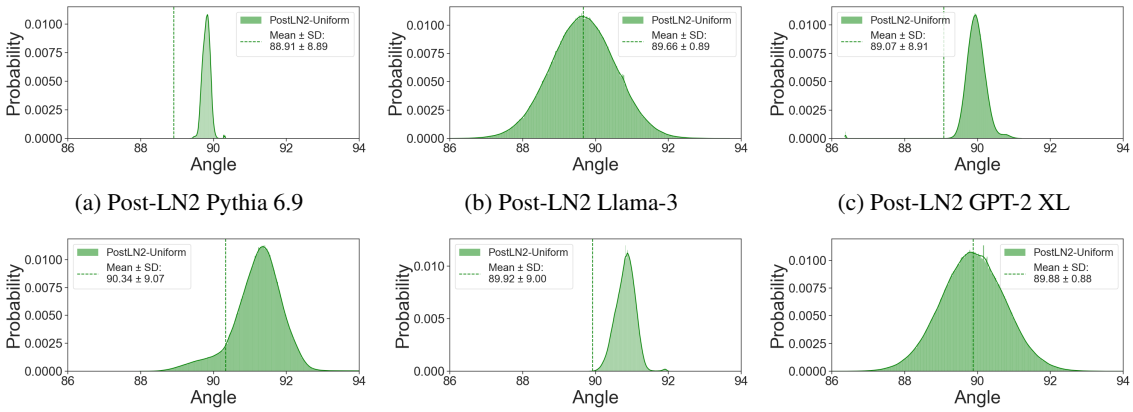
Figure 11: Distribution of angles (in degrees) between Hidden vectors (a-d) and post-normalization vectors (e-h) with the uniform vector for GPT-2 XL (Layer 43), GPT-Neo (Layer 18), Pythia 1.4 (Layer 18), and Llama-2 (Layer 24) for a specific layer chosen



(a) Pre-LN2 Pythia 6.9 (b) Pre-LN2 Llama-3 (c) Pre-LN2 GPT-2 XL

(d) Pre-LN2 GPT-Neo (e) Pre-LN2 Pythia 1.4 (f) Pre-LN2 Llama-2

Figure 12: Distribution of angles (in degrees) between (PreLN2 and Uniform) vectors for all models for a single layer chosen (Layer 43 for GPT-2 XL, Layer 18 for GPT-Neo and Pythia 1.4, and Layer 24 for Llama-2, Pythia 6.9, and Llama-3)



(a) Post-LN2 Pythia 6.9 (b) Post-LN2 Llama-3 (c) Post-LN2 GPT-2 XL

(d) Post-LN2 GPT-Neo (e) Post-LN2 Pythia 1.4 (f) Post-LN2 Llama-2

Figure 13: Distribution of angles (in degrees) between (PostLN2 and Uniform) vectors for all models for a single layer chosen (Layer 43 for GPT-2 XL, Layer 18 for GPT-Neo and Pythia 1.4, and Layer 24 for Llama-2, Pythia 6.9, and Llama-3)